

ls ---> used to list files and dirs

ls -l --> list files and dirs with long format

touch test1 ---> used to create empty file

ls -lt ---> recently created files will be displayed at beginning

ls -lrt --> recently created files will be displayed at the end

pwd ---> present working directory

mkdir dirname ----> create dir

cd dirname ----> change directory

cd dir/dir2/

cd .. --> used to come out of directory

cd ../../.. --->

mkdir -p temp1/temp2/temp3 ---> create complete directory structure

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cat filename ---> used to check content of a file

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vi is used to edit file

vi filename

esc --> i ---> insert

esc --> :wq! ---> save content of a file (write and quit)

esc ---> :q! --> quit without saving content

how do replace aprticular string or word in a file?

esc :%s/current-string/new-string/g

esc :%s/linux/windows/g

Need to replace only in 4th line

esc:4s/linux/windows/g

need to replace from 2nd to 4th line

esc:2,4s/linux/windows/g

how do you replace from 2nd line to end of a file

esc:2,\$s/linux/windows/g

how do set line numbers

esc : set nu --> set line numbers

esc : set nonu ---> remove line number

need to move cursor to particular line

esc :line_number

esc : 4 ---> cursor will move to 4th line

delete particular line in vi

esc press dd

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cp is used to copy file

cp file1 file2 ----> file1 will be copied to file2

cp test1 temp1/temp2/temp3/

copy dir to another dir

cp -r dir1 dir2

cp test1 file1 test3 temp ---> copy multiple files or dirs to another dir

how do rename file or dir?

mv file1 file2

mv file1 dir2

mv file1 file2 file3 dir1 ---> move multiple files to directory

install tree command

sudo yum install tree

rwxr-xr--
7 5 4

chmod is used to change permission of a file or dir

chmod 777 filename

chmod -R 777 dir

all subdirectories and files permission will be changed to 777

u--> user, g--> group and o--> others

chmod u+rwx filename

chown --> used change ownership of a file or a dir

chmod newowner filename

if new owner belongs to different group

chmod newowner:groupname filename

df -h . ----> used to check disk size

. indicates present disk

du -sh filename (disk usage)

used to check size of a file

du -sh * ---> all files and dirs size in present directory

echo "hi how r u" is used to print

echo -e "hi\nhow r u"

> (redirect)

used to store output of a command to file

ls -lrt > log

>> (append)

used to attach output of a command to end of a file

echo "this is linux class" >> log

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wc filename

lines words chars filename

wc -l log ---> number of lines

wc -w log ---> words

wc -c log --> chars

"head" is used to display 1st portion of a file

head -3 filename ---> 1st 3 lines

head -12 filename ---> 1st 12 lines of a file

head filename ---> 10 lines by default

tail is used to display last portion of a file

tail -3 filename --> last 3 lines

tail -5 filename ----> last 5 lines

tail filename -----> last 10lines by default

| (pipe) is used to give output of one command as input to next command

ls -lrt | wc -l

head -4 log | tail -1 --> display 4th line

how do you display 99th line of a file

head -99 file | tail -1

display 4th to 7th line

head -7 file | tail -3

count number of words in 99th line of a file

head -99 log | tail -1 | wc -w

display 2nd last line of a file

tail -2 log | head -1

grep --> pattern search

used to search string in file

grep "pattern" filename

grep -w "string" filename ----> search only specified word

grep -i "Linux" test_1 ---> i --> case insensitive

how do you search multiple pattern?

grep -e "linux" -e "windows" test_1

or
egrep

how do you list filenames if files have particular word or string

grep -R -l "linux" *

how do you print lines which don't have specified pattern
grep -v "linux" test_1

list lines if starts with particular pattern

grep "^pattern" filename

grep "s\$" test_1 ---> list lines ends with particular pattern

get command to print lines which start with "M" and ends with "s"

grep -c "linux" test_1 ---> count number lines which have specified pattern

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how do you replace string in a file?

sed --> used to replace string in a file
sed 's/linux/unix/g' filename

sed -i 's/linux/unix/g' test_1

sed '4s/linux/unix/g' ---> replace string in 4th line only

sed '2,6s/linux/unix/g' ---> from 2nd to 6th line

sed '\$s/linux/unix/g' ---> replace string only in the last line

delete particular line

sed '4d' test_1 --> delete 4th line of a file

sed '\$d' test_1 --> delete last line

sed '2,6 d' test_1 --> delete 2nd to 6th line

sed '1d;3d' test_1

how do you print particular line?

sed -n '3p' test_1 ---> print 3rd line

sed -n '\$p' test_1 ---> print last line

sed -n '2,6p' test_1 ---> print 2nd to 6th line

sed -n '2p;6p' test_1 ---> print 2nd to 6th line

cut --> used to cut file colounwise

cut -d " " -f1 data --> 1st coloumn

cut -d " " f1,3 data ---> 1st and 3rd coloumn

cut -d " " f3-6 data ---> 3rd to 6th column

awk is used to cut file rowwise and coloumn wise but we use awk command to cut file coloumn wise

awk -F " " '{print \$1}' data ----> 1st coloumn

\$1 ---> 1st coloumn

\$2 --> 2nd coloumn

\$NF ---> last coloumn

\$(NF-1) --> 2nd last coloumn

awk -F " " '{print \$NF}' data ---> last coloumn

awk -F " " '{print \$(NF-1)}' data --> 2nd last coloumn

find ---> used to find location of a file

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find . -iname "filename"
```

i --> case insensitive

list files which are modified 3months back

```
find . -type f -mtime +90
```

```
find . -type d -mtime +90
```

```
find . -mtime +90 -->both files and dirs
```

list files which are modified within 3months

```
find . -type d -mtime -90
```

find files which are modified with in 50mins

```
find . -type f -mmin -50
```

find files which modified 50mins ago

```
find . -type f -mmin +50
```

find files which have 777 permission

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find . -type f -perm 777
```

find all empty files

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find . -type f -empty
```

-----commnad to list all non empty files-----

```
find . -maxdepth 1 -iname "test1"
```

maxdepth is used to retriect automatic recursive

maxdepth 1 ---> 1st level dir

maxdepth 2 ---> 2nd level dir

delete all files which are modified 3 months ago

```
find . -tyep f -mtime +90 | xargs rm -rf
```

xargs is used to pass output of one command to next command as arguements

Assigment:

1. command to list files if their size is more than 1MB

2. use -exec option in find command to delete files

softlink

it's shortcut to a file. if you make any changes in the actual file, it will get reflected in the softlink

if you delete actual file, softlink will not work

```
ln -s /home/ec2-user/file1 link1
```

hardlink

```
ln /home/ec2-user/file1 hard1
```

```

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diff b/w hard link and softlink

if I delete actual file, softlink will not work
if I delete actual file, hardlink works fine because it points to inode of a
file (inode is uniq number indicates memory address)

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Assignment:
1. command to list files if their size is more than 1MB

2. use -exec option in find command to delete files

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ps --> used list current running processes on the system

need to check whether particular process is running or not

ps -ef | grep "processname"

3.need to list all processes started by particular user

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stop process (forceful stop)
kill -9 PID

4.kill all process started by particular user
5. kill all process by name

you can use sleep command for practice

sleep 10
sleep 100
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how do stop process gracefully

sudo service name stop
sudo service name start
sudo service name restart

6. get systemctl command

sudo systemctl stop name
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umask is used to set default permission on system.
it should run with root user

umask 000 ---> whoever creates file or dir will have full permission

umask 777 ---> no permission

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rwxr-xr-x = 755

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777
755

022

umask 0222

r-xr-xr-- = 554

777
554

223

umask 233

su (switch user or superuser)
is used to login as other user or root user

su - username
password:

sudo (superuser does)

sudo ---> used to execute command with root permission

sudo yum install vim wget

add user ABC abd and give sudo permission

sudo useradd ABC
sudo passwd ABC
password:
password:

sudo su - (login as root user from ec2-user)

vi /etc/sudoers.d/90-cloud-init-users

go to last line of this file

give same permission for ABC also like ec2-user
you need to add another line and change ec2-user to ABC
save file
ABC also will have sudo permission

more command is used to display content of a file page wise

more filename

less ---> used to display content of a file pagewise, but you can scroll up and scroll down using up and down arrows

less filename

(press q to quit)

date --> used to print system date and time
date +%m

uniq --> print onlu uniq data and removes deplicates. use this along with sort
sort --> used to sort data --> ascending sort
sort -r filename ---> decending sort

cat test | sort | uniq

we need to use uniq command always with sort command

how do you check linux version

uname -v ---> only version

uname -a ---> all info

who ---> who all logged into system
whoami ---> check ur username
hostname --> hostname or ip of current server

ssh --> used to login to remote server and uses port 22

ssh username@server
password:

ssh -i aws-keys/devops-june-21.pem ec2-user@172.31.21.0

you need to private IP

run command on remote server

ssh -i aws-keys/devops-june-21.pem ec2-user@172.31.21.0 "ls"

1. install winscp
 2. using winscp copy pem file to server1
 3. login to server1 using putty
 4. change permission to 400 using chmod 400 filename.pem
 5. ssh -i filename.pem ec2-user@server2-private-ip
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scp is used to copy files from one server to another (uses port 22)

scp filename username@server:/path/
password: